AWS CLOUD NETWORKING

VPC Peering Implementation Report

Connecting two isolated Amazon Virtual Private Clouds

Field	Details
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AWS Region	Asia Pacific (Mumbai) — ap-south-1

1. Project Overview

This project demonstrates private network communication between two Amazon Virtual Private Clouds (VPCs) in the AWS Mumbai Region using VPC Peering. A peering connection provides private IPv4 communication between resources in the two VPCs when the required routes and security-group rules are configured.

2. Network Architecture and Configuration

The implementation uses two VPCs with non-overlapping IPv4 CIDR ranges.

Component	Name	CIDR / ID	Purpose
VPC A	ewb-vpc-a	10.10.0.0/16 vpc-0bfc34571c1caa708	Requester VPC
Public subnet A	ewb-a-public	subnet-073a5420d83bbc16a	Bastion host
Private subnet A	ewb-a-private	subnet-0a1ba06dcf15bfa69	App-A instance
VPC B	ewb-vpc-b	10.20.0.0/16 vpc-092aca3d0a2153edf	Accepter VPC
Private subnet B	ewb-b-private	subnet-0f2836a19f548ab46	App-B instance
Internet gateway	(attached to VPC A)	igw-0a3df4a0d040b2fd9	Internet access for bastion

The VPC peering connection is named **ewb-a-b-peer** and has connection ID **pcx-06aee6250bb7cc03a**. Both VPCs belong to the same AWS account (751446514374) in ap-south-1. The AWS console screenshot records its status as **Active**.

3. Routing Configuration

The route tables contain local routes for their own VPC CIDR and peering routes for the remote VPC CIDR. The public route table in VPC A also contains a default route to the Internet Gateway for the bastion host.

Route table	Destination	Target
ewb-rt-a-private	10.10.0.0/16	local
ewb-rt-a-private	10.20.0.0/16	pcx-06aee6250bb7cc03a
ewb-rt-b-private	10.20.0.0/16	local
ewb-rt-b-private	10.10.0.0/16	pcx-06aee6250bb7cc03a
ewb-rt-a-public	10.10.0.0/16	local
ewb-rt-a-public	10.20.0.0/16	pcx-06aee6250bb7cc03a
ewb-rt-a-public	0.0.0.0/0	igw-0a3df4a0d040b2fd9 (Internet Gateway)

4. Security Group Configuration

Security groups were configured to permit SSH access (TCP port 22) to the instances and ICMP traffic for connectivity testing between the private instances.

Security group	Group ID	Inbound rules
ewb-sg-bastion	sg-070af0dec5f3d3c23	SSH / TCP 22
ewb-sg-app-a	sg-084f2028cd4880964	SSH / TCP 22; All ICMP - IPv4
ewb-sg-app-b	sg-096fb35cab2e0ee19	SSH / TCP 22; Custom ICMP - IPv4 (Echo Request)

5. Instance Access and Connectivity Tests

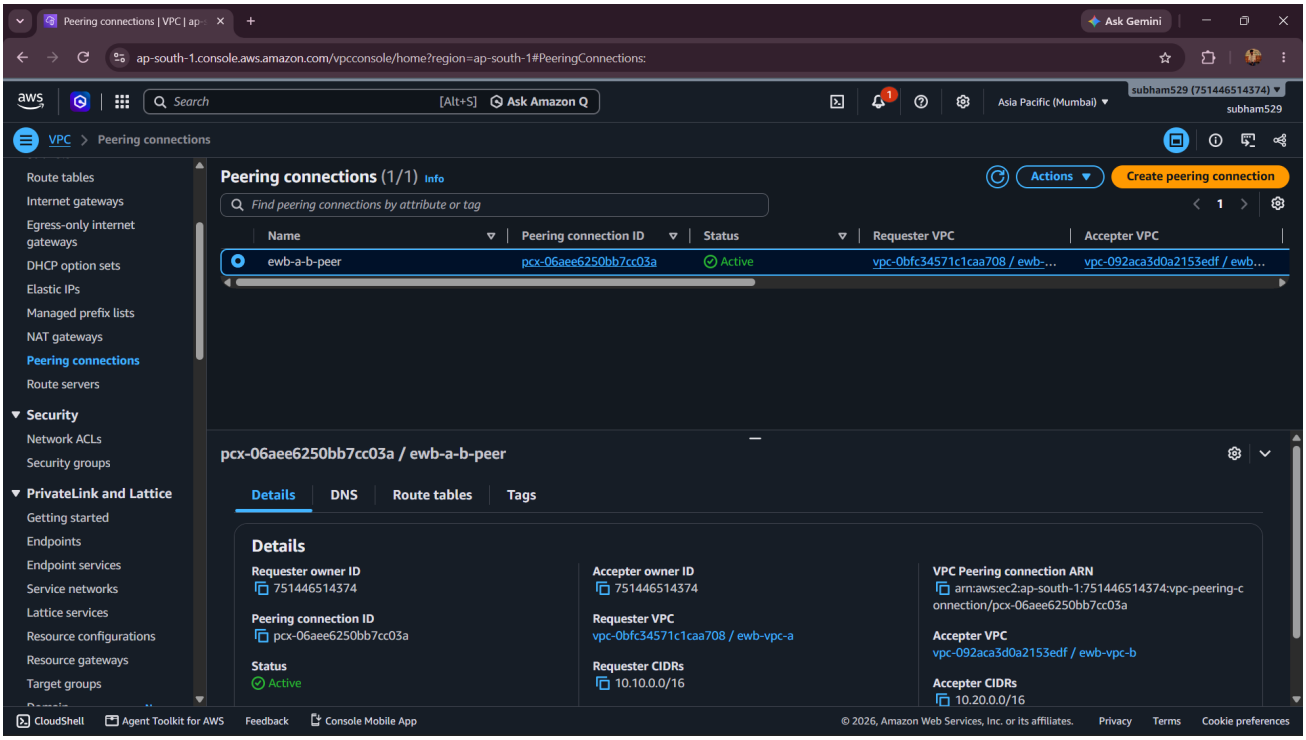
The bastion host in the public subnet of VPC A is the controlled SSH entry point. The private instances (App-A in VPC A and App-B in VPC B) are reached through the bastion using an SSH ProxyCommand/ProxyJump-style connection, so neither private instance needs a public IPv4 address.

This report documents the network configuration (peering connection, routes and security groups) through AWS console evidence. Terminal ping results are not included in this report.

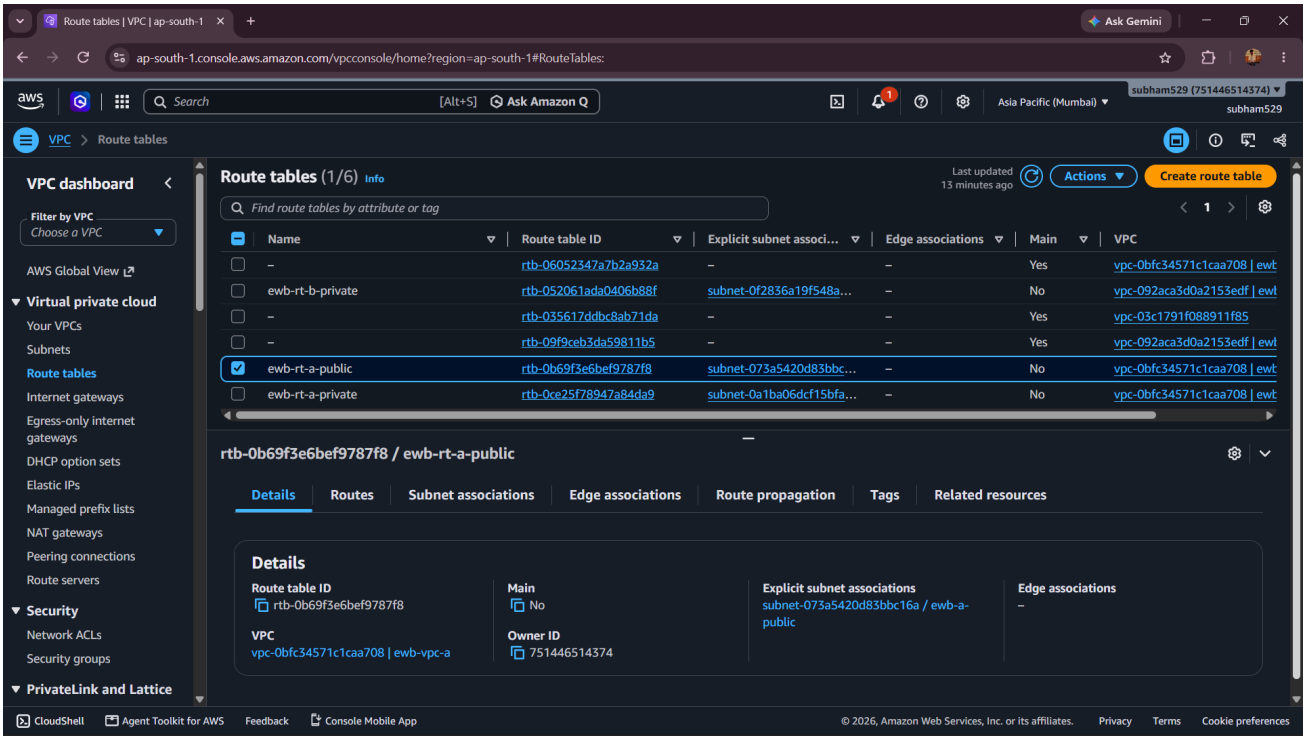
6. Evidence Screenshots

The screenshots below are included once each. Similar console captures have been selected to avoid repeating the same evidence.

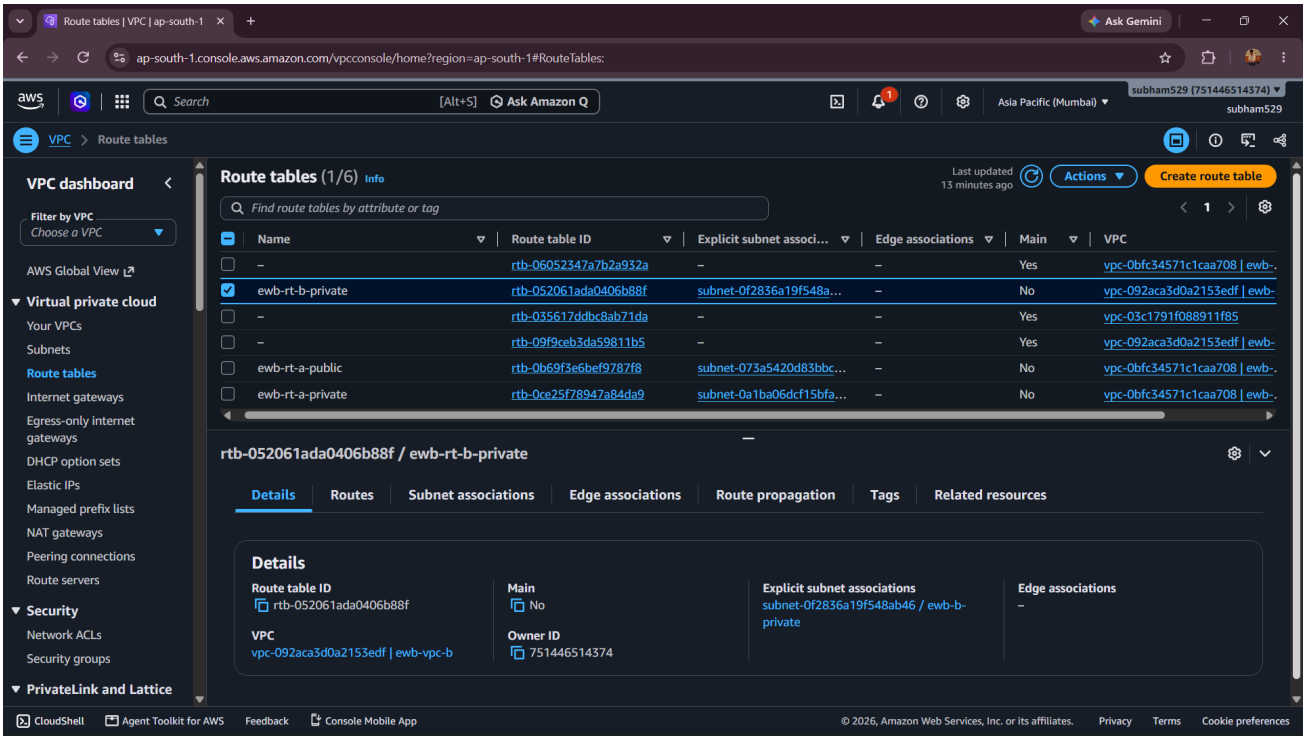
Evidence 1. VPC peering connection ewb-a-b-peer shown as Active.



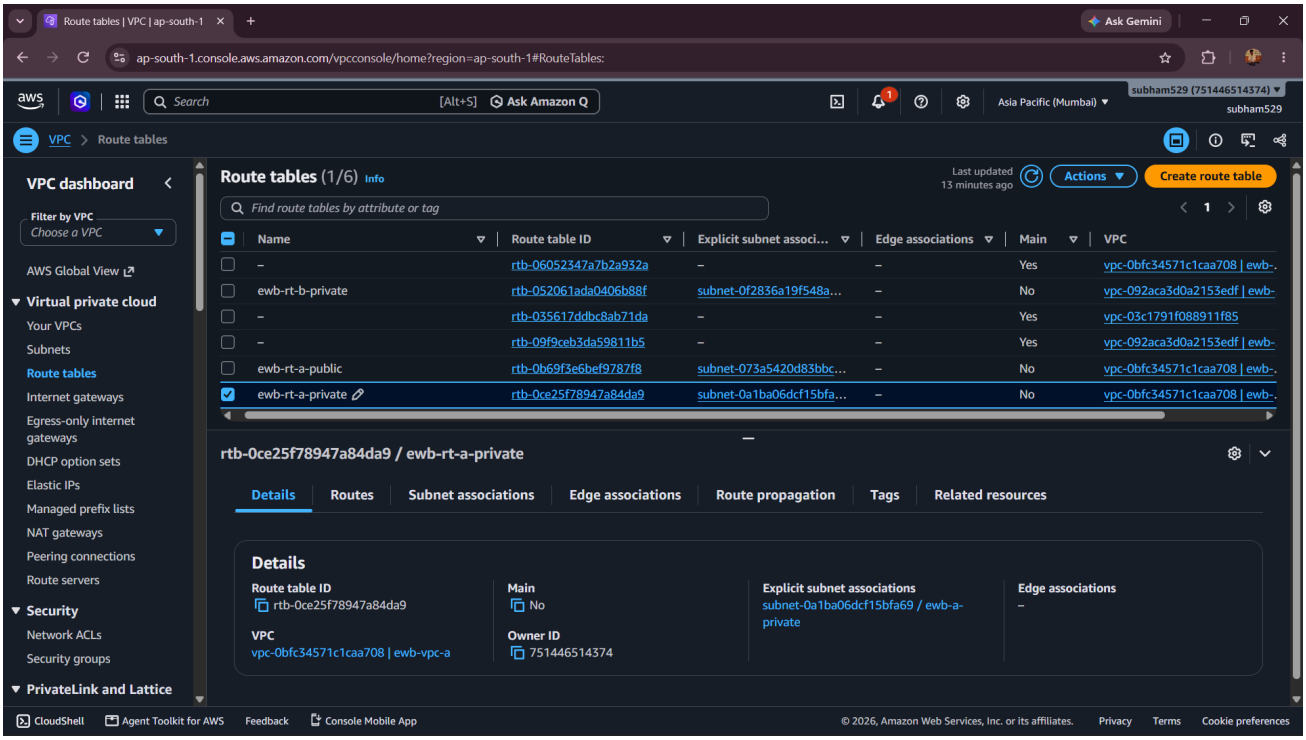
Evidence 2. Route table details for ewb-rt-a-public.



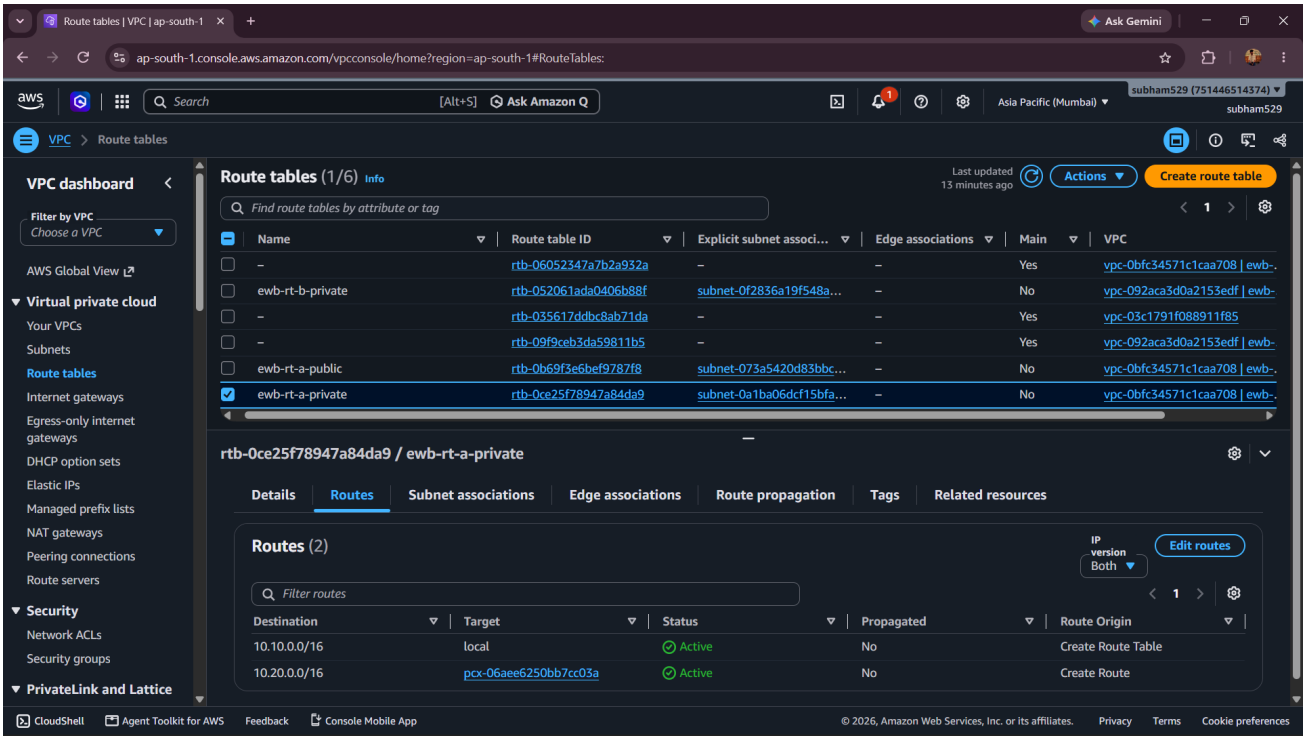
Evidence 3. Route table details for ewb-rt-b-private.



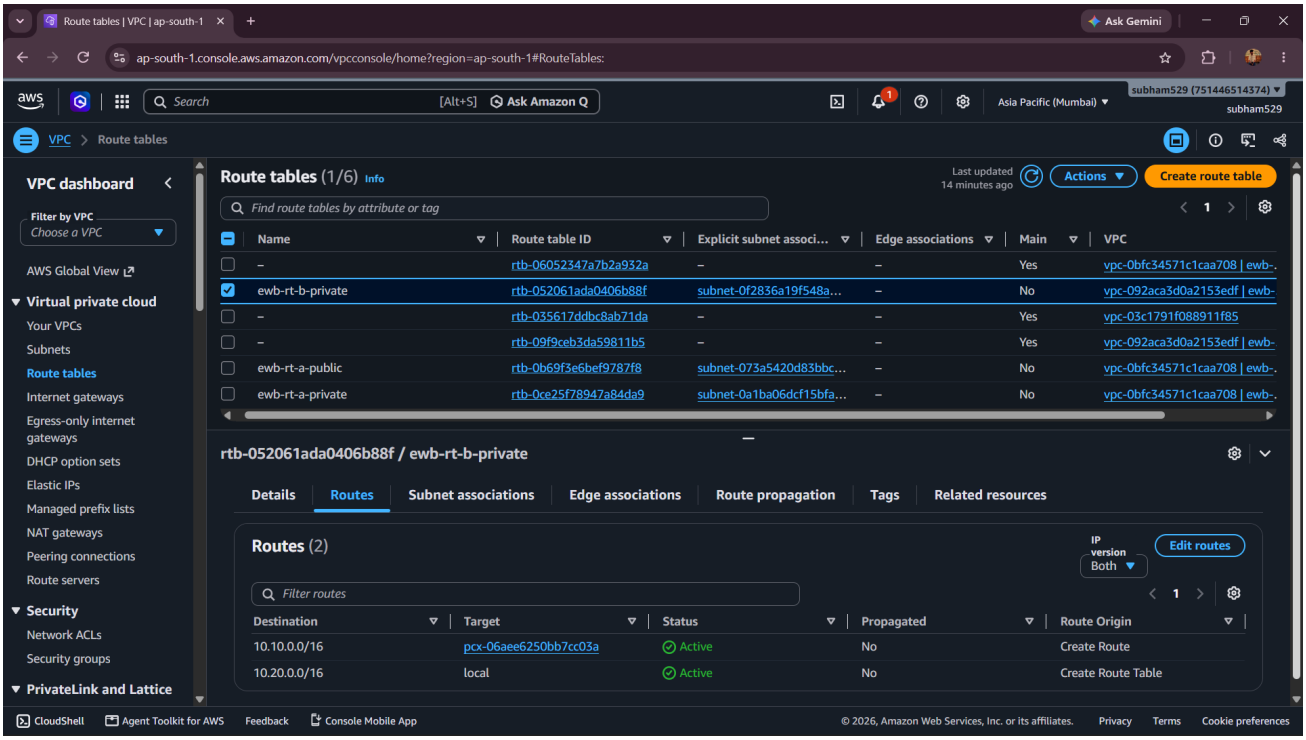
Evidence 4. Route table details for ewb-rt-a-private.



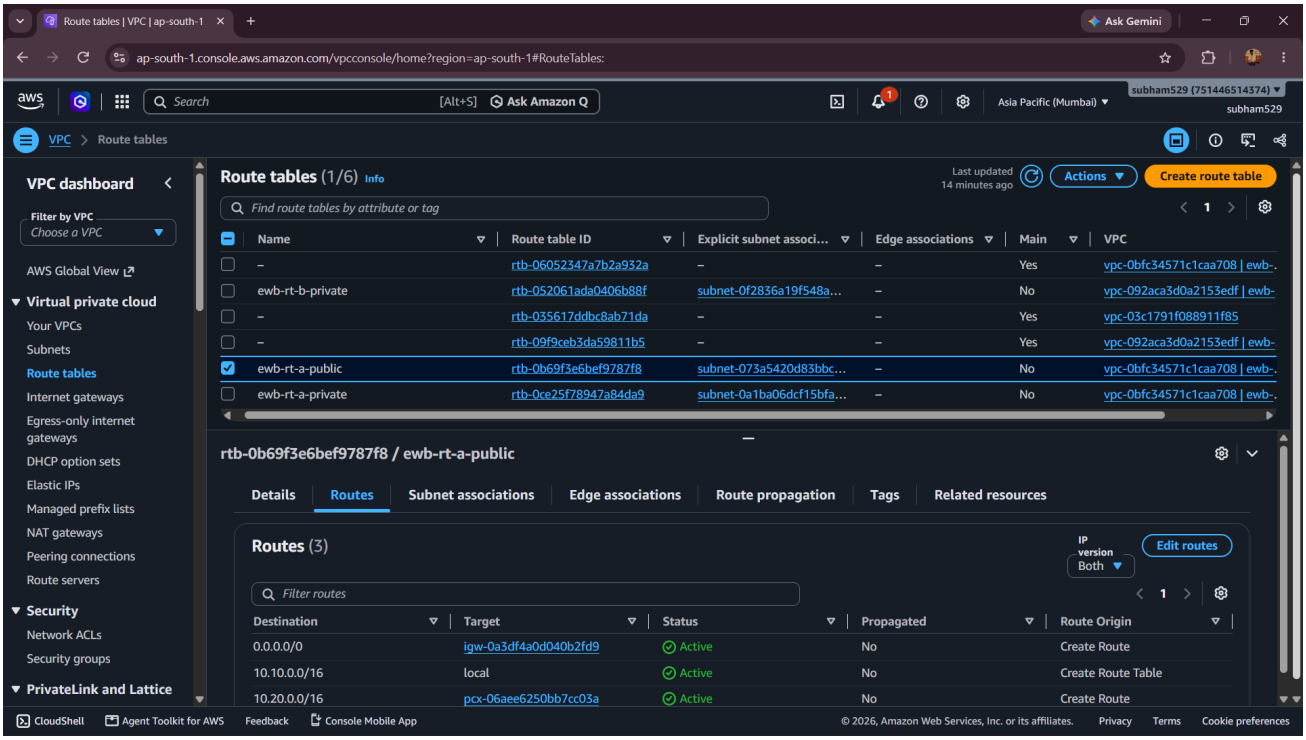
Evidence 5. VPC A private route table with local and peering routes.



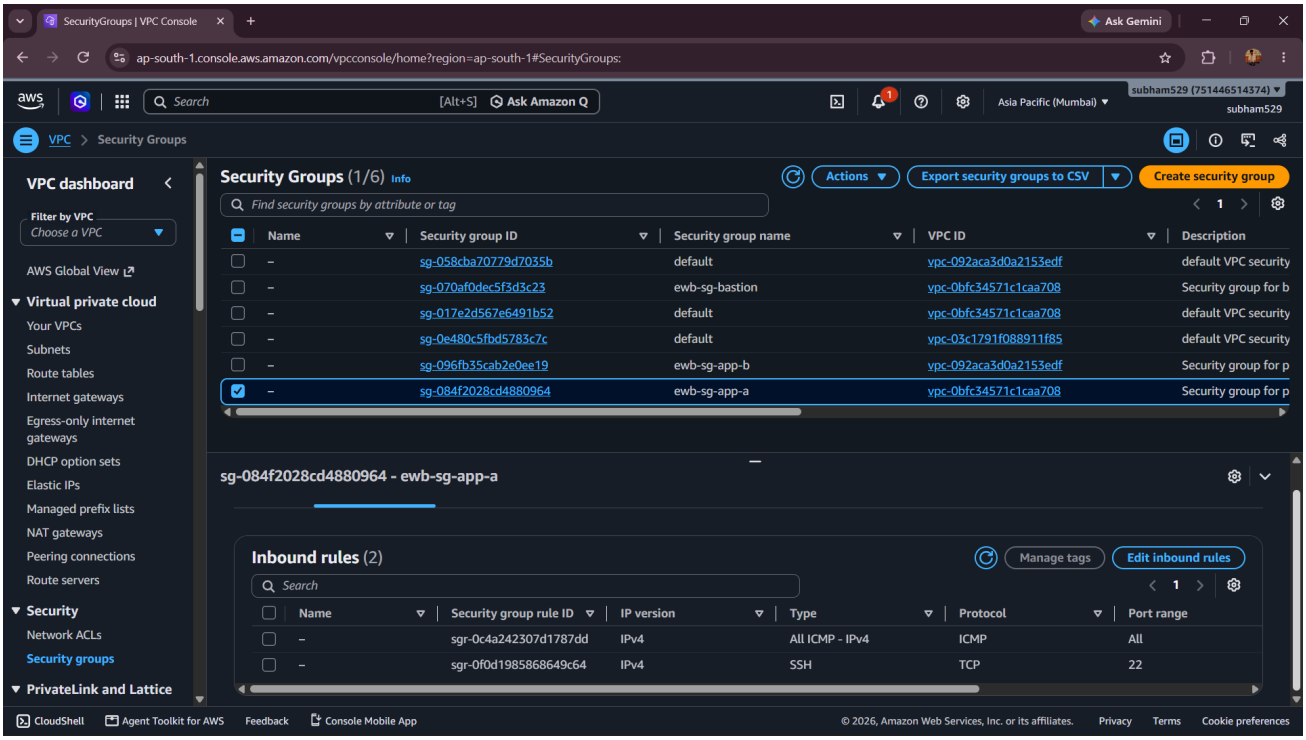
Evidence 6. VPC B private route table with local and peering routes.



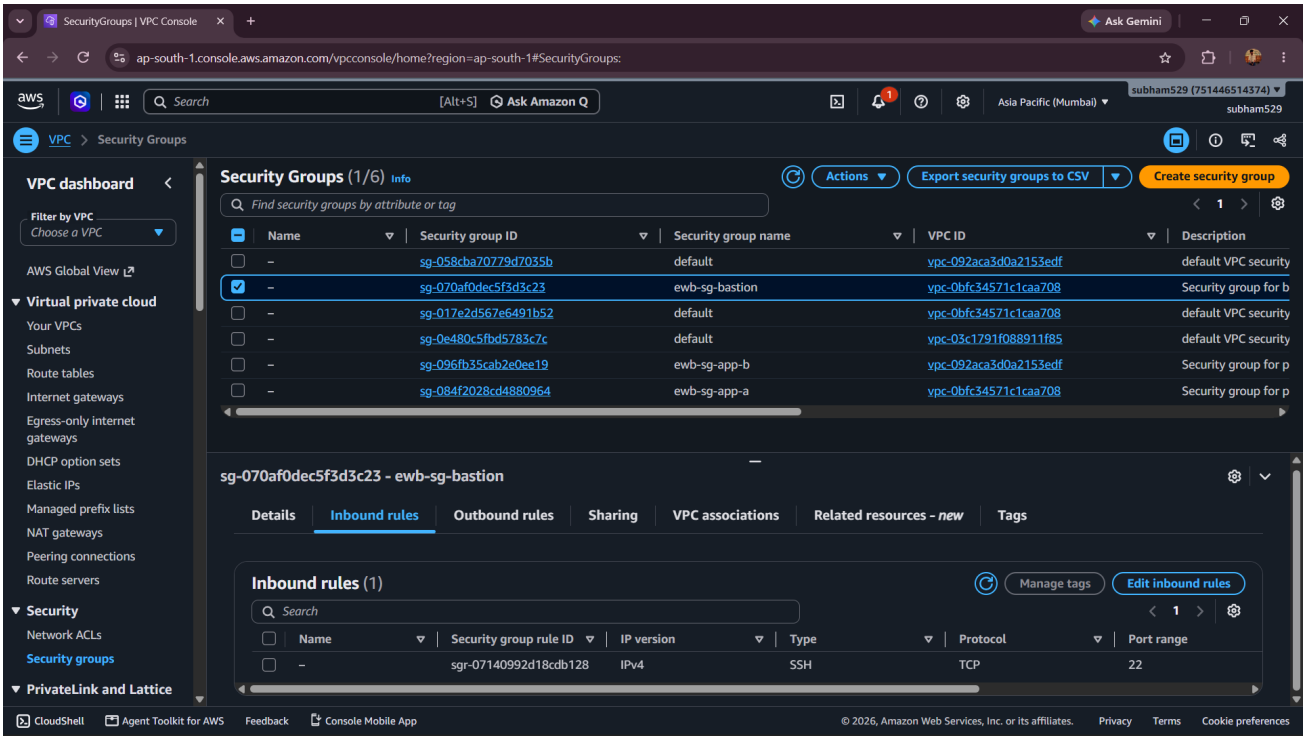
Evidence 7. VPC A public route table with Internet Gateway and peering routes.



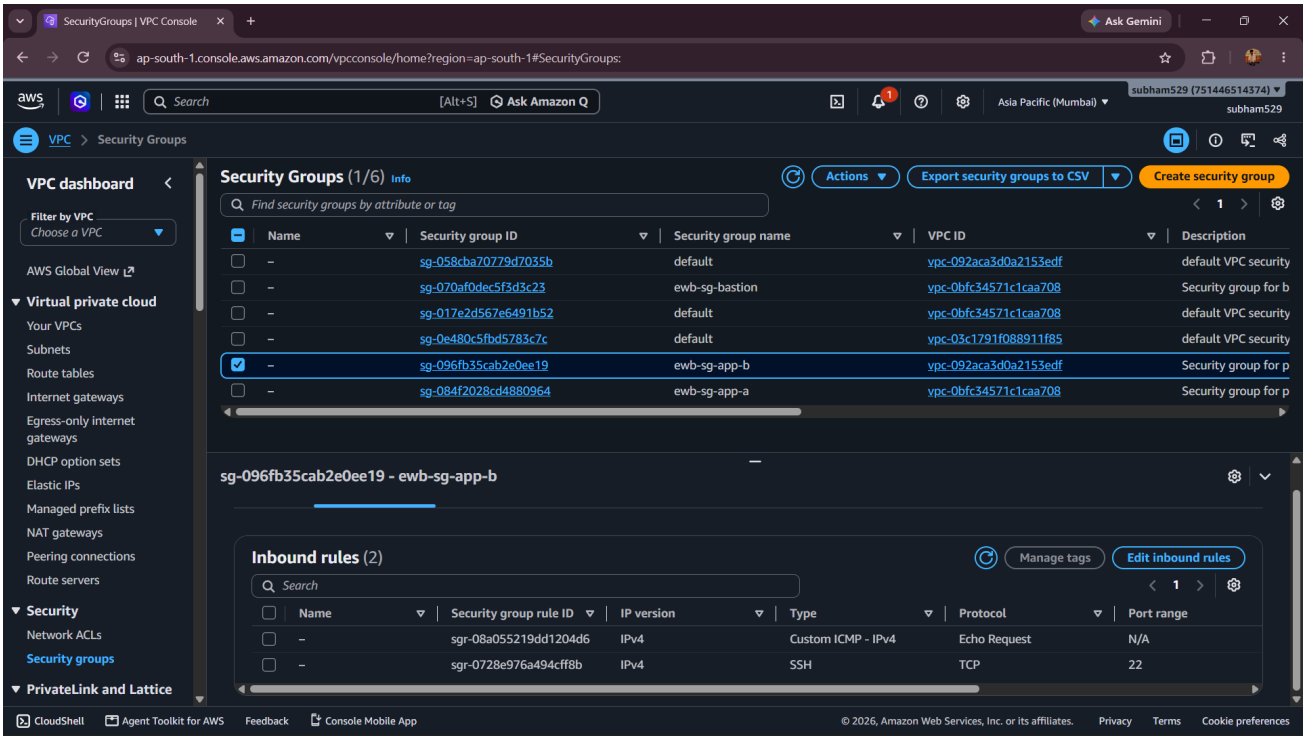
Evidence 8. Inbound rules for ewb-sg-app-a.



Evidence 9. Inbound SSH rule for ewb-sg-bastion.



Evidence 10. Inbound rules for ewb-sg-app-b.



7. Conclusion

The AWS environment demonstrates the configuration of two VPCs (10.10.0.0/16 and 10.20.0.0/16), an active VPC peering connection, route-table entries on both sides for private communication, and security-group rules for controlled SSH and ICMP traffic. These settings provide the foundation for private communication between the two VPCs through VPC Peering.